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Administration

▼ Relevant source files

`admin/utilities/bulkEmailController.php``admin/utilities/bulkEmailMain.php``admin/utilities/bulkEmailSystem.php``admin/utilities/writeIndexablePagePerDB.php``admin/verification/checkXHTML.php``admin/verification/longOperationInit.php``admin/verification/orderLinksByTitle.php``admin/verification/rebuildCalculatedFields.php``admin/verification/rebuildRecordTitles.php``admin/verification/verifyFieldTypes.php``admin/verification/verifyValue.php``hclient/widgets/profile/profile_edit.html``hclient/widgets/profile/profile_edit.js``hclient/widgets/viewers/3dViewer.php``hclient/widgets/viewers/3dhopViewer.php``hserv/utilities/DbUtils.php``hserv/utilities/DbVerify.php``hserv/utilities/USystem.php``startup/index.php`

This document covers Heurist's administrative tools and systems for database management, user administration, system verification, and bulk operations. These tools are primarily intended for

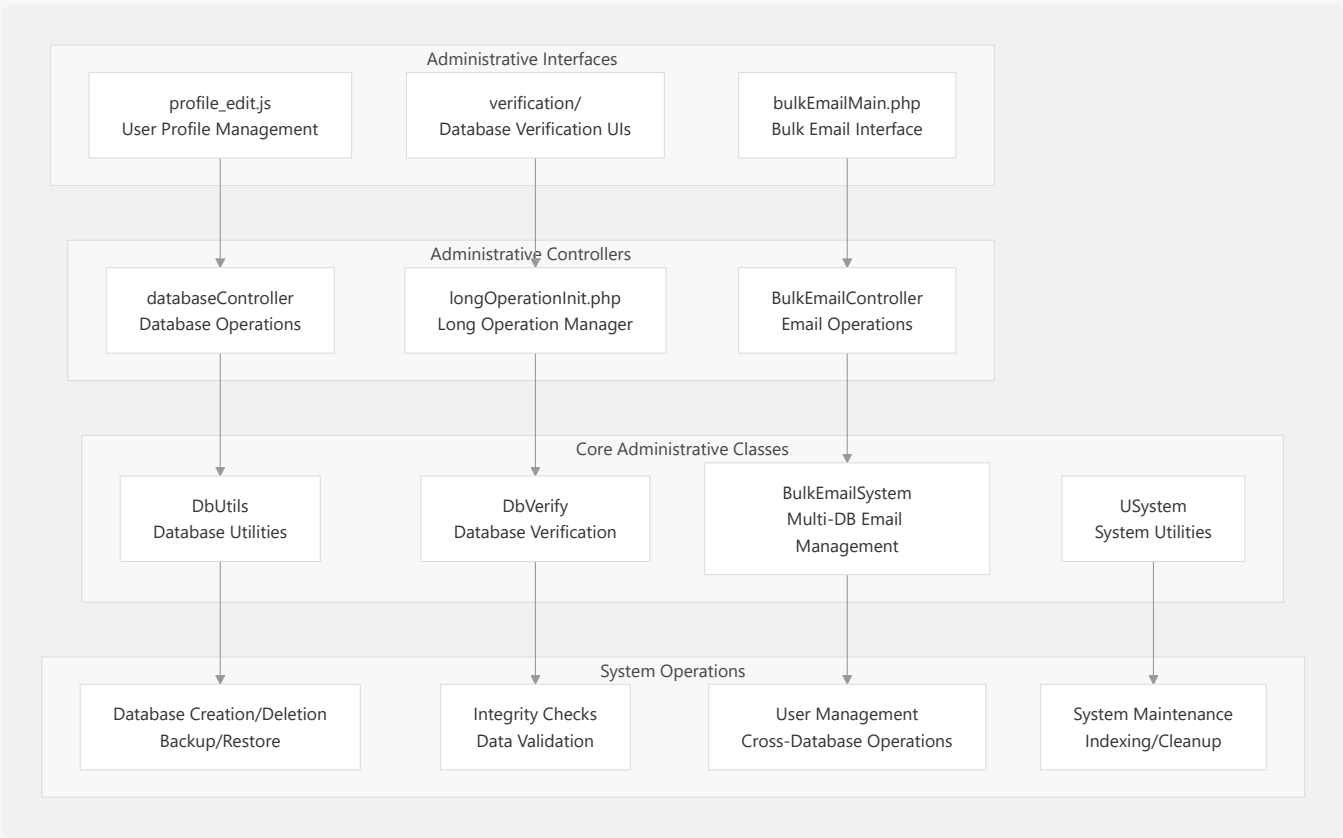
system administrators and database managers who need to maintain Heurist installations, manage users across multiple databases, and perform system-wide operations.

For database-specific administration tools, see [Database Administration](#). For user management functionality, see [User Management](#).

Administrative System Overview

Heurist's administration system provides comprehensive tools for managing multi-database installations, performing system maintenance, and coordinating operations across multiple databases on a server.

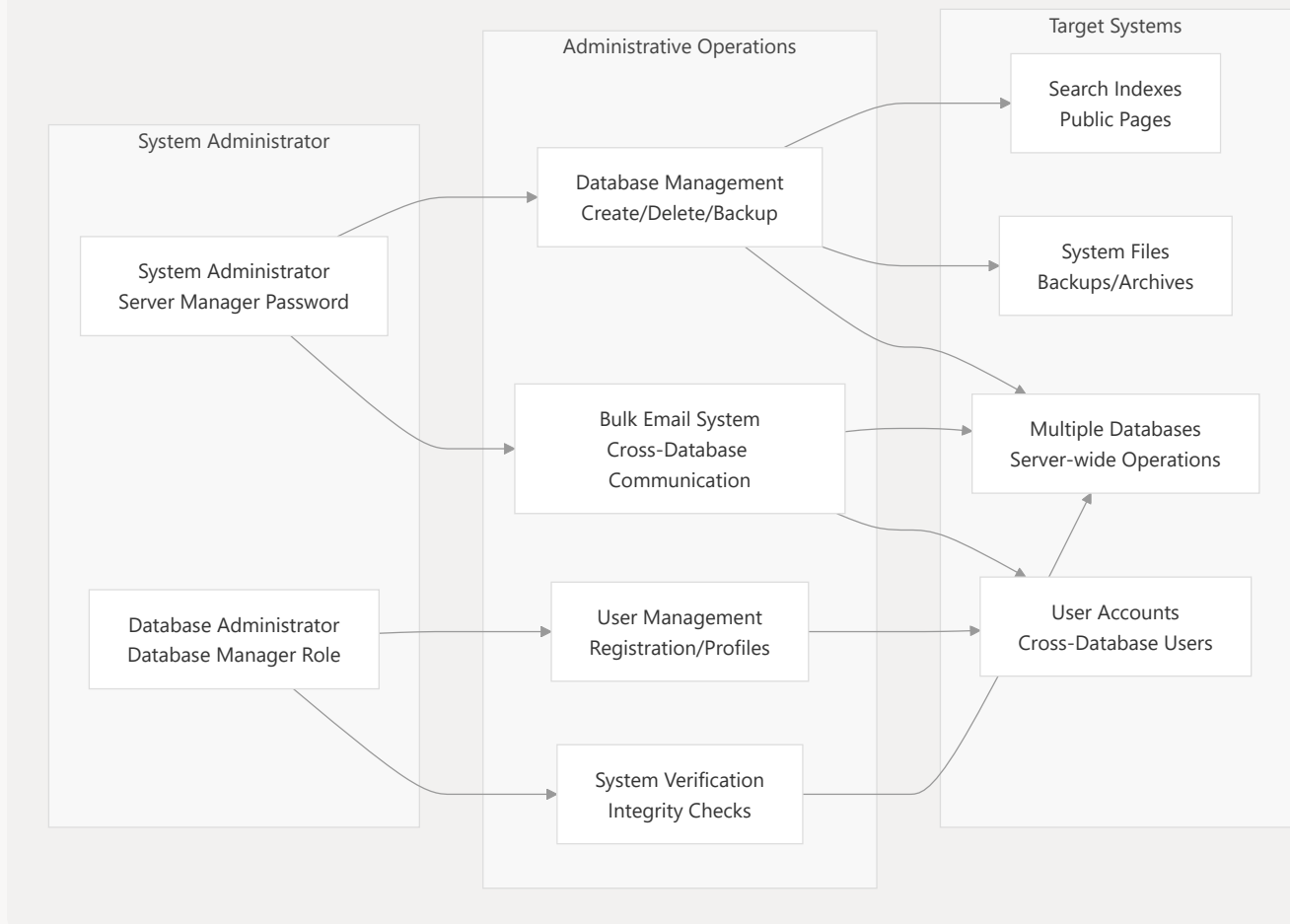
Core Administrative Architecture



Sources: [admin/utilities/bulkEmailSystem.php](#) 1-1467 [admin/utilities/bulkEmailMain.php](#) 1-1054

[hserv/utilities/DbUtils.php](#) 1-2000 [hserv/utilities/DbVerify.php](#) 1-3856

Administrative Data Flow



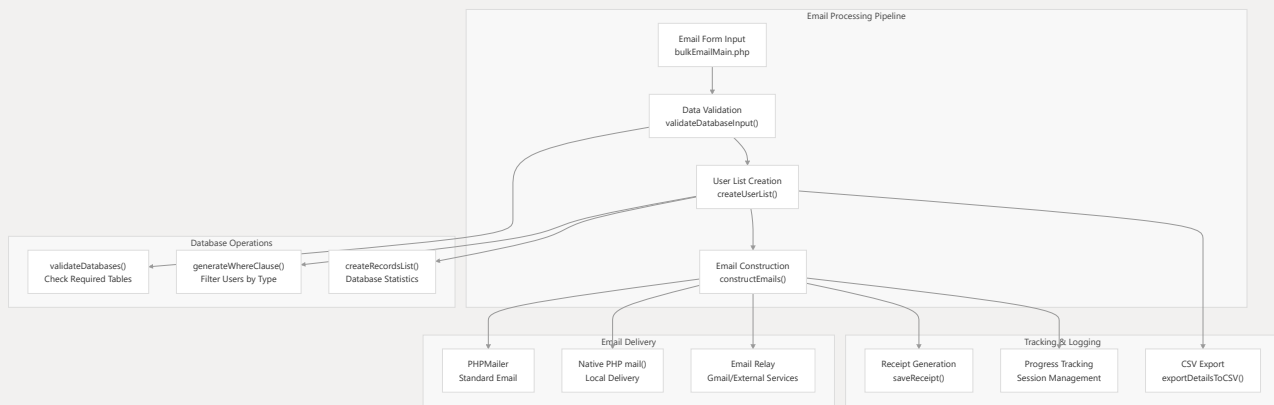
Sources: `admin/utilities/bulkEmailController.php` 32-674 `hserv/utilities/USystem.php` 29-690
`admin/utilities/writeIndexablePagePerDB.php` 1-679

Bulk Email System

The bulk email system enables system administrators to send announcements and communications to users across multiple databases on a Heurist server.

Email System Architecture

The `BulkEmailSystem` class provides comprehensive functionality for managing bulk email operations:



Key email system methods include:

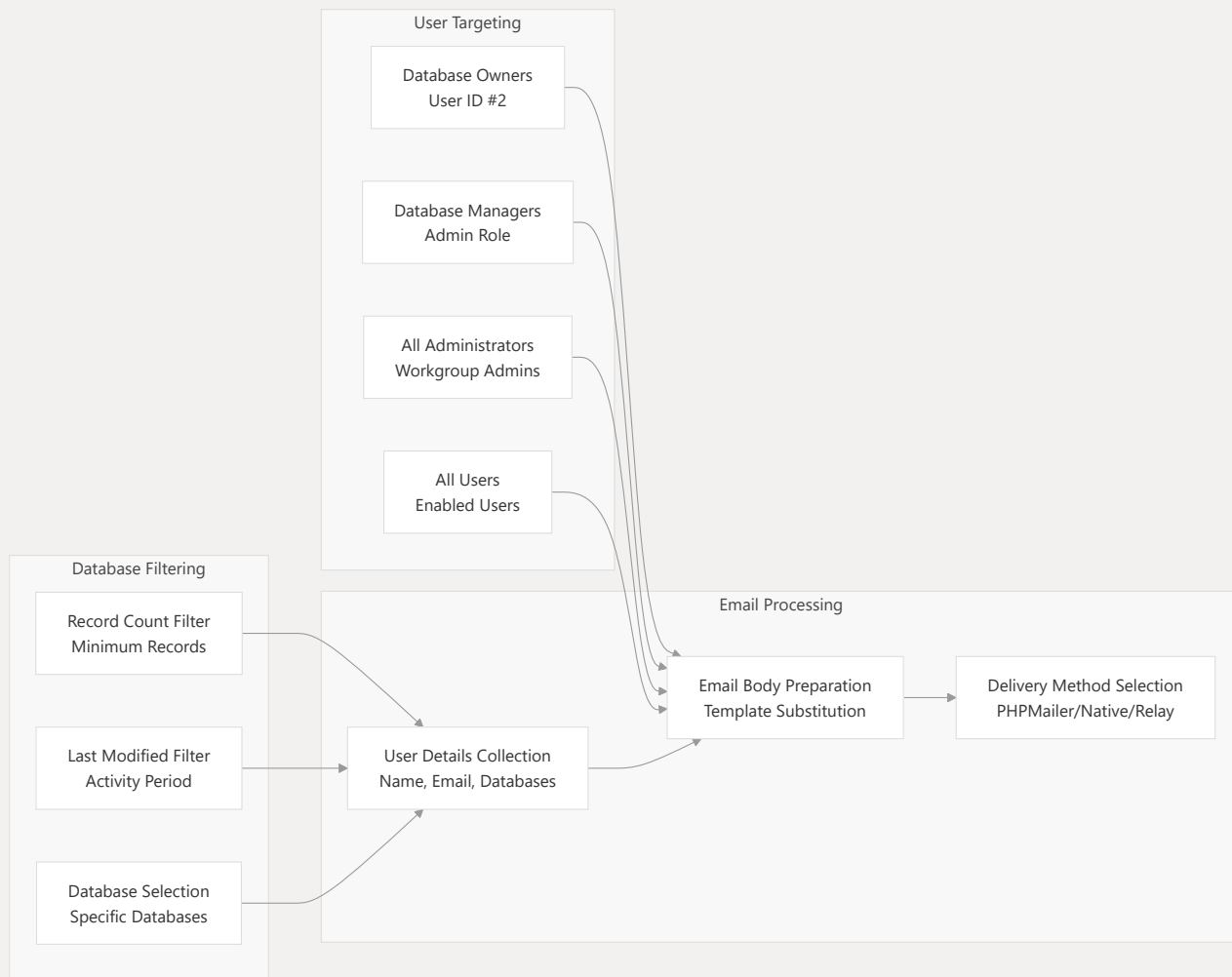
- **processFormData()** - Validates and processes form input
- **createUserList()** - Builds lists of users across databases
- **constructEmails()** - Handles email composition and delivery
- **exportDetailsToCSV()** - Exports user details for analysis

The system supports email template substitution with placeholders like **##firstname##**, **##database##**, and **##records##**.

Sources: `admin/utilities/bulkEmailSystem.php` 63-1467

`admin/utilities/bulkEmailMain.php` 32-1054

Email Filtering and Targeting



The system includes sophisticated filtering based on database activity, user roles, and database characteristics.

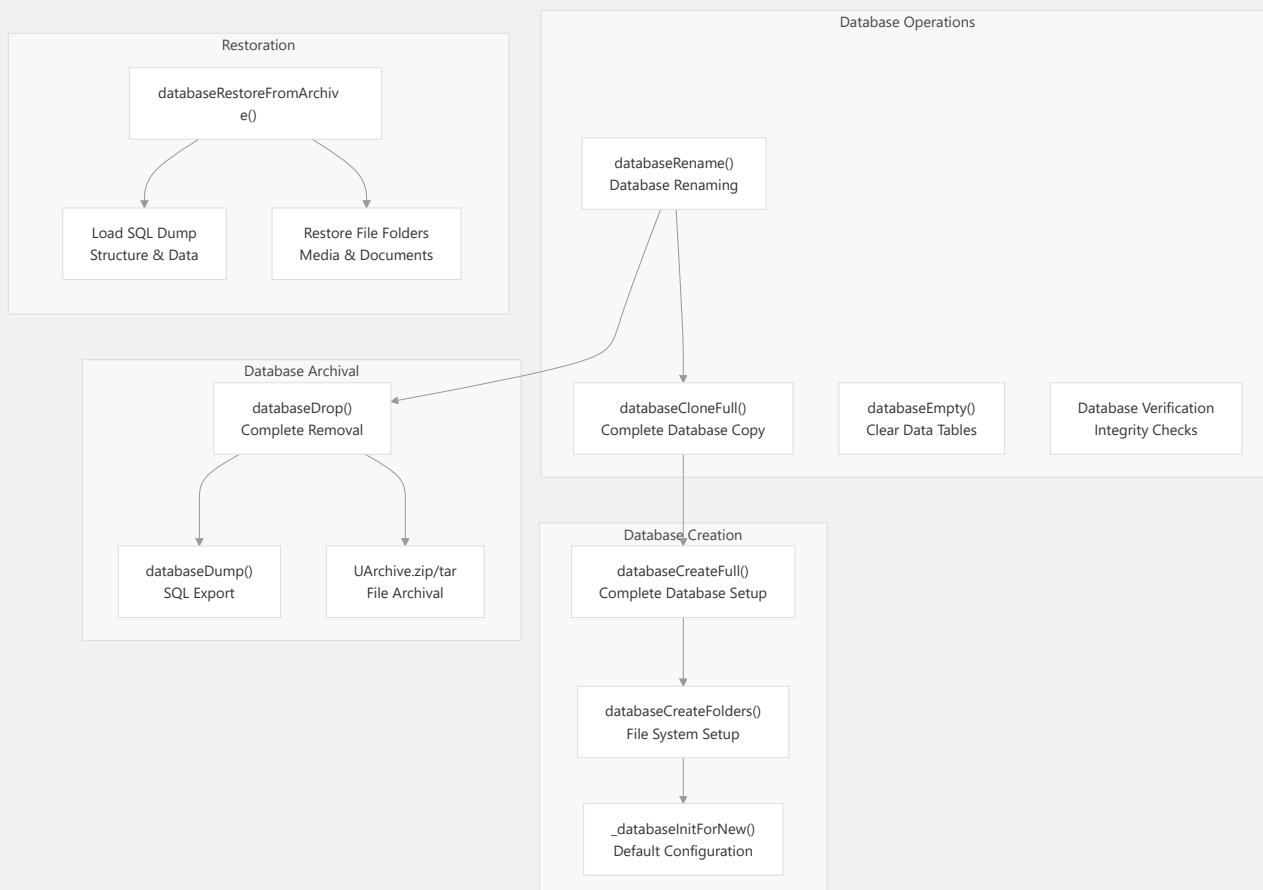
Sources: `admin/utilities/bulkEmailSystem.php` 434-505

`admin/utilities/bulkEmailController.php` 67-674

Database Management Operations

Database management in Heurist centers around the `DbUtils` class, which provides comprehensive database lifecycle management.

Database Lifecycle Management

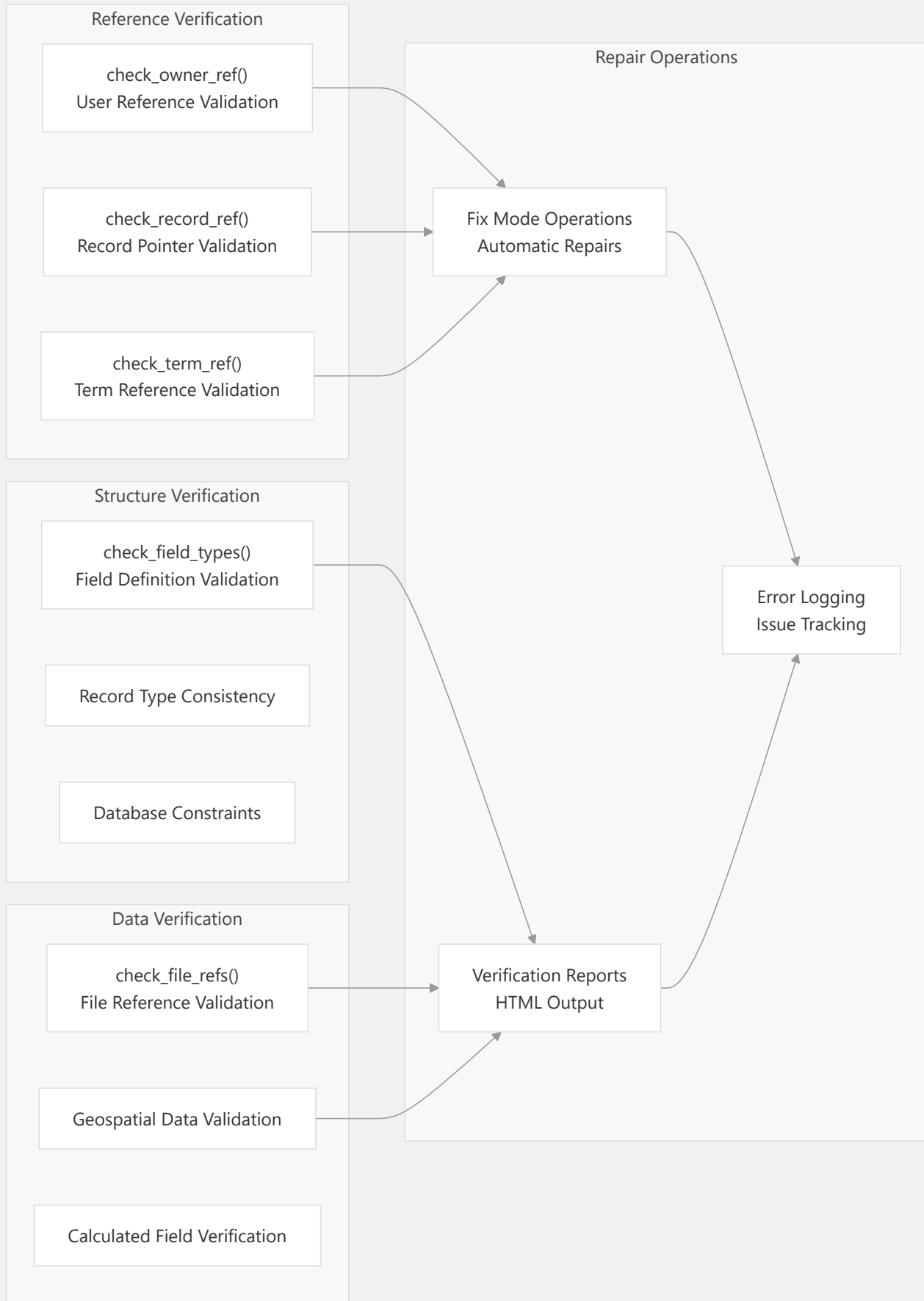


The **DbUtils** class manages database operations with proper error handling and progress tracking.

Sources: `hserv/utilities/DbUtils.php` 61-2000

Database Verification System

The **DbVerify** class provides comprehensive database integrity checking:



Key verification methods include:

- `check_owner_ref()` - Validates user ownership references

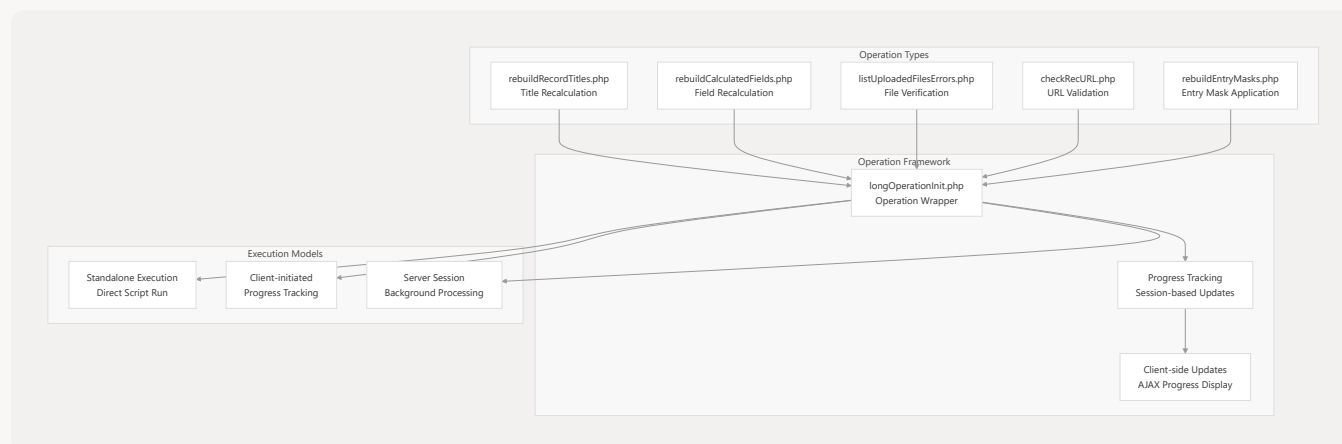
- `check_record_ref()` - Verifies record pointer integrity
- `check_term_ref()` - Validates vocabulary term references
- `check_file_refs()` - Checks file and media references

Sources: `hserv/utilities/DbVerify.php` 32-3856 `admin/verification/verifyValue.php` 31-295

System Verification Tools

Heurist includes multiple verification and maintenance tools for ensuring system integrity.

Long Operation Management



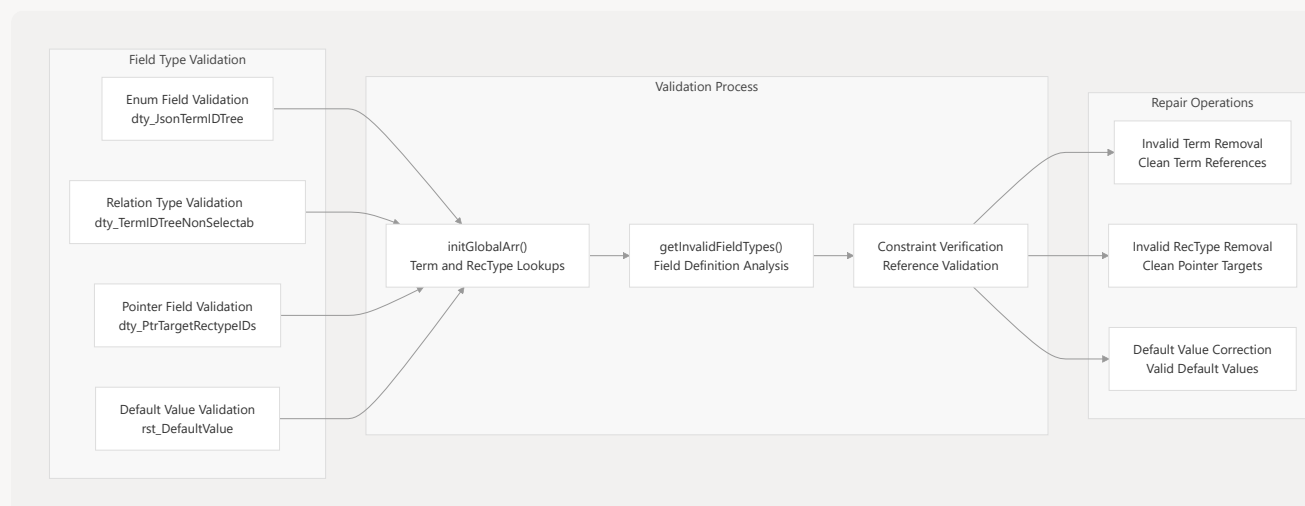
The verification system supports both immediate execution and progress-tracked operations for large databases.

Sources: `admin/verification/longOperationInit.php` 1-112

`admin/verification/rebuildRecordTitles.php` 1-252

`admin/verification/rebuildCalculatedFields.php` 1-186

Field Type Verification



The field type verification system ensures that all field definitions maintain valid references to